Susmita Sett
susmitasett04@gmail.com

Employment

- **Telescope Operator and data analyst** (2025-Present) - Commonwealth Scientific and Industrial Research Organisation (CSIRO) / Square Kilometer Array Observatory (SKAO)
- **Associate Lecturer, Level A** (2023-2025) - Curtin Institute of Radio Astronomy, Curtin University, Bentley, WA - Research and Teaching

Education

- **Curtin University**, Bentley, WA 2020–2023
  PhD in Astronomy and Astrophysics
  Thesis: Exploring the southern pulsar population in image domain with the MWA
- **The University of Manchester**, Manchester, UK 2017–2020
  MSc by Research, Astronomy and Astrophysics
  Thesis: Search for radio pulsars in supernova remnants
  Grade: **First, Merit**
- **Christ University**, Bangalore, India 2014–2017
  BSc Physics, Mathematics and Chemistry
  Percentage: **84%**

Teaching and Supervision Experience

- **Introduction to Astronomy, ASTRO1003** - Designed and delivered lectures, incorporating modern pedagogical approaches and updated course materials to enhance student engagement and learning outcomes. Developed and implemented curriculum updates aligned with current research in astronomy and astrophysics.
- **PhD student co-supervision** - Angira Mahida, Curtin University (2024-Present): Searching for intermediate-mass black holes in globular clusters.
- **Undergraduate and honours student co-supervision** - Thomas Campbell, Curtin University (2024-Present): Combining the capabilities of optical and radio astronomy.

Research Experience

- **Postdoctoral research: Studying X-ray binaries and transients using the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST)**
  - Exploring LSST's capability for early detection of outbursts in low-mass X-ray binaries
  - Warped accretion disks and superorbital period recovery with LSST
  - Prospects of detecting intermediate mass black hole tidal disruption events using LSST
  - Paving the pathway to future: combining the capabilities of SKAO and LSST
- **PhD project: Exploring the southern sky population of pulsars using image-based methodologies with the MWA.**
  - Developed and commissioned a MWA-VCS specific imaging pipeline on Pawsey Supercomputing Systems such as Garrawarla, Setonix as early adopters of the system.
  - Developed methodologies to detect pulsar candidates in continuum images based on pulsar properties.
  - Candidates followed up using MWA-VCS data, GMRT or higher-frequency telescopes like Parkes.
  - Conducted image-based surveys of the Galactic plane to detect new pulsar candidates.
  - Performed population and spectral studies of low-frequency pulsars to explore emission mechanisms.

- **Breakthrough Listen project: Casual Research Assistant**
  - Assisted the Breakthrough Listen organization with observations using the Parkes Radio Telescope.
  - Focused on searching for signals of extraterrestrial activity and other interesting objects in the Galaxy.
- **Masters thesis project: Search for radio pulsars in supernova remnants**
  - Searched for radio pulsars in supernova remnants using high-time-resolution data from the Green Bank Telescope (GBT).
  - Processed and analyzed data using PRESTO and Python.
  - Achieved 2 to 16 times higher sensitivity compared to previous surveys, though no new pulsars were detected.
  - Published first-authored work in the Astronomy and Astrophysics journal (A&A).
- **Undergraduate summer project: Data analysis of gravitational waves**
  - Extracted signals from noise in gravitational wave detectors like LIGO and VIRGO.
  - Participated in the IndiGo mock data challenge to understand gravitational wave data analysis.
- **Undergraduate summer project: Reduction and extraction of stellar spectra of Be stars**
  - Analyzed spectral and emission lines of B[e] stars under the guidance of Dr. K.T. Paul.
  - Utilized image reduction software (IRAF), gnuplot, and Python for analysis.

PUBLICATIONS

1. Warped accretion disks and Superorbital period recovery with the Vera C. Rubin Observatory's Legacy Survey of Space and Time (2025)
   S. Sett, K. Dage, A. Bahramian, S.C. Fu et al. - Submitted to Publications of the Astronomical Society of Australia (PASA).
2. Exploring LSST's capability for early detection of outbursts in low-mass X-ray binaries (2025).
   S. Sett, A. Bahramian, K. Dage et al. - Accepted for publication in Publications of the Astronomical Society of Australia (PASA).
3. Imaging pulsar census of the Galactic Plane using MWA VCS data (2024).
   S. Sett, M. Sokolowski, E. Lenc and R. Bhat - Accepted for publication in Publications of the Astronomical Society of Australia (PASA).
4. Image-based searches for pulsar candidates using MWA VCS data (2023).
   S. Sett, R. Bhat, M. Sokolowski and E. Lenc - Accepted for publication in Publications of the Astronomical Society of Australia (PASA).
5. No evidence for accretion around the intermediate-mass black hole in Omega Centauri (2025).
   A. Mahida, **S. Sett** et al - Accepted for publication The Astrophysical Journal.
6. Investigating four new candidate redback pulsars discovered in the image plane (2025).
   F. Petrou, **S. Sett** et al - Accepted for publication in Publications of the Astronomical Society of Australia (PASA).
7. Classifying compact radio emission in nearby galaxies: A 10 GHz study of active galactic nuclei, supernovae, anomalous microwave emission, and star-forming regions (2025)
   K. Dage, **S. Sett** et al - Accepted for publication in The Astronomical Journal.
8. Radio Continuum studies of ultracompact and short orbital period X-ray binaries (2025)
   K. Dage, **S. Sett** et al - Accepted for publication in The Astrophysical Journal
9. Detecting the black hole candidate population in M51's young massive star clusters: constraints on accreting intermediate-mass black holes (2025)
   K. Dage, **S. Sett** et al - Accepted for publication in The Astrophysical Journal
10. Single-pulse analysis and average emission characteristics of PSR J1820-0427 from observations made with the MWA and uGMRT (2023).
    P. Janagal, M. Chakraborty, N. D. R. Bhat, S. J. McSweeney and **S. Sett** - Accepted for publication in Monthly Notices of the Royal Astronomical Society (MNRAS).
11. Hyperbolic limit on the early arrival time of bright pulses from PSR J0835-4510 (Vela) (2023).
    Angiraben D. Mahida, J. L. Palfreyman,G . Molera Calvés, **S.Sett** - Accepted for publication in Monthly Notices of the Royal Astronomical Society (MNRAS).

12. Search for radio pulsars in five nearby supernova remnants (2021).
S. Sett, R. P. Breton, C. J. Clark, M. H. Kerkwijk and D. L. Kaplan - Accepted for publication in Astronomy & Astrophysics (A&A)

Recent talks

- **Indian Institute of Astrophysics**, Bangalore, India — November 2024
  Invited Talk
- **Raman Research Institute**, Bangalore, India — November 2024
  Invited Talk
- **Commonwealth Scientific and Industrial Research Organisation**, Kensington, Perth, WA — June 2024
  Invited Talk
- **MWA Project Meeting**, Online and in person — 2021, 2022, 2024, 2025
- **Astronomical Society of Australia Annual Science Meeting**, Online — 2021, 2024

Honours and Scholarships

- **Women in STEM Grant (AUD 10,000) —** Co-investigator for a Government funded proposal supporting STEM initiatives in Western Australia.
- **Program for Early Academic Careers (PEAC) —** Selected participant in a competitive program for emerging academic leaders.
- **Research Training Program (RTP) Scholarship —** Australian Government scholarship awarded for outstanding research potential (PhD).
- **CSIRO Top-Up Scholarship —** Competitive award for high-achieving PhD candidates in relevant research areas.

Accepted proposals

- **Parkes Radio Telescope**
  - Timing and spectral investigation of PSR J1002-2036 - 22 hours — 2022
  - Broadband follow-up of PSRs J1357-2530 and J0452-3418 - 8 hours — 2021
  - Searching for pulsars in four supernova remnant - 7.5 hours — 2020
- **Giant Metrewave Radio Telescope (GMRT)**
  - GMRT follow up of six MWA pulsar candidates - 7 hours — 2023
  - Probing the plasma turbulence toward high Galactic latitudes - 20 hours — 2022
  - GMRT followup of PSR J0036-1033 - 16 hours — 2022
- **Murchison Widefield Array (MWA)**
  - MWA shadowing LSST: A multi-wavelength view of the transient sky - 65 hours — 2026
  - SKA-Low station sensitivity measurements using pulsar co-observations with the MWA - 12 hours — 2024

International Collaborations and Professional Societies

- The Square Kilometer Array Observatory
- The Legacy Survey of Space & Time - Stars, Milky Way and Local Volume - Transient and Variable Stars Working Groups.
- ASKAP Variables and Slow Transients.
- Murchison Widefield Array Collaboration.
- MWA Pulsars and Fast Transients Group (MWA-PFT).

- MWA Transients Collaboration.
- The Southern-sky MWA Rapid Two-metre (SMART) pulsar collaboration
- Astronomical Society of Australia (ASA).

EDUCATION AND OUTREACH

- **Judging panel at Kids in Space WA** 2026
  Organised by the Andy Thomas Space Foundation.
- **Invited panel as part of the School Summit- Inspiring Future Careers** 2026
  Organised by Adjuvant and Murdoch University, WA.
- **Invited talk and panel as part of the Summer Space Series** 2026
  Organised by University of Adelaide and the Andy Thomas Space Foundation.
- **Previous chair of CIRA Development Committee (DevCom)** 2024-2025
  Committee promoting an inclusive environment where all can thrive.
- **Media Contact, Vera C. Rubin Observatory** 2025
  Selected for scientific outreach and public engagement.
- **Organising committee for ICRAR-Con** 2024
  ICRAR-Con is the annual gathering of everyone from the two nodes of ICRAR for recap and future planning.
- **Co-organiser of ICRAR Student Day** 2020
  A one day conference for students and staff associated with ICRAR.
- **Volunteer and speaker at AstroFest** 2020-2024
  A Perth-based astronomy outreach activity organized by ICRAR.
- **Participating in event, 'Women in HPC'** 2022
  An interview and blog to encourage careers of women in high performance computing sectors
  `https://www.linkedin.com/feed/update/urn:li:activity:7041347707537379328/`
- **Public talk on Pulsars** at Bolton School, Bolton, UK 2019
  Outreach talk for the high school students of the Astronomy club of the school.
- **Exhibition explainer and presenter** 2017-2019
  Jodrell Bank Discovery Centre, Manchester, UK

CERTIFICATIONS

- The X-series programme in astrophysics, conducted by the Australian National University for securing 100 % in the exam and assignments of the courses. The X-series programme comprised of four courses conducted by Dr. Paul Francis and Dr. Brian Schmidt.
- Introduction to Cosmology, conducted by the MP Birla Institute of Fundamental Research, Bangalore, India. This course was conducted by C. Sivaram, Professor Emeritus of the Indian Institute of Astrophysics.
- General relativity and Cosmology, a course conducted by Professor Sivaram, Indian Institute of Astrophysics, Bangalore, India
- Neutron Stars and Black Holes, course conducted by Dr. G. Srinivasan.
- Project Management by Google.

TECHNICAL SKILLS

- **Radio Astronomy Software tools**
  - WSCLEAN: Proficient in performing wide-field imaging using this software.
  - CASA: Comprehensive experience in data processing and analysis using the Common Astronomy Software Applications package. I have experience in using CASA for data reduction for GMRT and ATCA imaging data.
  - CARTA, DS9 and Aladin: Extensively used these softwares to visualise and analyse radio images as part of my PhD work.
- **Time-domain specific software**
  - PRESTO: Used for pulsar search and analysis.
  - DSPSR: Utilized for real-time pulsar signal processing.
  - PSRCHIVE: Proficient in archiving and analyzing pulsar data.
  - PAZI: Expertise in pulsar data processing and analysis.
  - PSRPOPPY: Skilled in simulating pulsar populations.
- **Telescope operation**
  - Currently part of the operation teams for the SKAO.
  - Have scheduled and performed observations with the Parkes (Murriyang) Radio Telescope for 2 years and ongoing as part of multiple projects.
  - Trained Australian Telescope Compact Array (ATCA) observer.
  - Trained Parkes Telescope observing expert.
- **Programming languages**
  - Proficient in Python : Capable of creating data-reduction and analysis pipelines from scratch for analysis of radio astronomy data from single dish and interferometric telescopes.
  - Also capable of parallelising such pipelines within framework of large computing cluster using the most efficient methods.
  - Shell scripting: Experienced in automating tasks and managing workflows using shell scripts.
  - Git: Proficient in version control and collaborative development using Git.
- **Others**
  - IRAF: Used for the reduction of optical and infrared data. I used it as part of a summer project during my undergraduate studies to deal with data for Be stars.
  - TOPCAT: Skilled to use TOPCAT for handling large survey catalogs.
  - HPC : Experienced in working and mitigating challenges faced with HPC systems. First-adopters and testers for Pawsey Supercomputing Systems such as Garrawarla and Setonix.
  - Telescope specific data processing for MWA, GBT, Parkes, GMRT and ATCA radio telescopes.